

3. Review The IoT and AI in Agriculture: The Time Is Now—A Systematic Review of Smart Sensing Technologies

The integration of the Internet of Things (IoT) and Artificial Intelligence (AI) is transforming modern agriculture by enabling precision farming, real-time monitoring, and data-driven decision-making. This systematic review analyzes recent developments in smart sensing technologies used in arable crops and grasslands, highlighting the growing adoption of IoT-based sensor networks and AI-powered analytics. Technologies such as optical, acoustic, electromagnetic, soil, and water sensors are widely used to collect real-time agricultural data, while machine learning models including Support Vector Machines (SVM), Convolutional Neural Networks (CNN), Random Forests, and Deep Learning algorithms help optimize irrigation, fertilization, crop monitoring, and disease detection. The review identifies several challenges, including high implementation costs, poor connectivity in rural areas, interoperability issues, cybersecurity risks, and concerns regarding data privacy and ownership. Emerging technologies such as Edge AI, blockchain-based data management, and autonomous agricultural robots are presented as promising solutions for improving efficiency, transparency, and sustainability. The study concludes that although IoT and AI have significant potential to revolutionize agriculture and enhance food security, broader adoption will require affordable technologies, supportive policies, improved digital infrastructure, and ethical governance to ensure that the benefits of smart farming are accessible to farmers of all scales.